Zhang Bihan

Shanghai, China | Email: Zhangbihan621@outlook.com | Website: bhannzhang.github.io

EDUCATION

Tongji University Master **Shanghai, China**
Interaction Design, concentration in human-computer interaction and cognition Sep. 2018 - Jun. 2021

- GPA: 4.73/5.0 (ranked 1st)
- Outstanding Graduate of Tongji University (top 2%), 2021

Hohai University Bachelor ***National A-rated program (top 10%) in China*** **Nanjing, China**
Electrical Engineering and Automation Sep. 2012 - Jun. 2016

- GPA: 4.08/5.0

AWARDS & HONORS

- **MUSE Design Awards, Silver (2024); American Good Design, Gold (2024)**
- **Outstanding Master's Thesis Award, Ton**gji University (sole recipient, **2021***)*
- **Hongda Scholarship(2018 ,2019) & Excellent Master's Scholarship (2020), Tongji University (Top 2-3%)**
- **National Third Prize, UXPA China | UXDA International User Experience Innovation Competition (2019)**
- **2018 National College New Energy Vehicle Big Data Innovation & Entrepreneurship Competition –** Entrepreneurship Group, National Second Prize

ACADEMIC RESEARCH EXPERIENCE

Principal Investigator (Student PI) – Mitsubishi Collaboration Project

- Led research on social acceptance and external human-vehicle interaction for L4+ autonomous campus vehicles, spanning interaction concepts and UX evaluation.
- **The work contributed to a first-authored journal article and an HHME 2019 conference paper.**

Team Lead – FAW Group Hongqi Collaboration Project

- Led a team to survey and model ~500,000 m² of real-world geography, developing a digital-twin simulation environment adopted as the core evaluation platform for virtual driving tests.

Principal Investigator (Student PI) – Shanghai Auto-Industry Foundation Project

- Investigated driver behavior under ACC/LKA; designed 10 driving scenarios and collected and analyzed behavioral data from 30 participants.

CarIXD & User Experience Lab, Tongji University

Principal Investigator (Student PI) – Baidu Collaboration Project

- Benchmarked 26 automotive brands and authored a 12,000-word white paper that translated findings into UI/UX guidelines for intelligent-cockpit interaction.
- Presented the findings at Baidu Headquarters; the work was recognized by Apollo Lab for methodological rigor.

Team Lead – Intelligent In-Vehicle Interaction System Environment

- Led a 6-member team in building the lab's first functional Unity driving-simulation platform, subsequently used in more than 30 HMI behavior tests.

SELECTED INDEPENDENT RESEARCH & PROTOTYPES

- **OYI: Embodied Companion Robot for Social Interaction:** Built a 3D-printed robot with articulated arms, voice interaction, persistent memory, and light feedback; conducted a five-day mixed-method study with 32 participants. ***(2026)***
- **Embodied Swarm Interaction:** Built an ESP32/IMU controller and a 55–60-agent Processing simulation; ran an n=6 pilot on embodied influence, mental-model formation, and collective legibility. ***(2026–Present)***
- **Movement as a Language:** Built a Processing research tool with 12 controlled motion probes; ran and coded an n=6 pilot on perceived state, intention, and agency. ***(2026–Present)***
- **Pneumatic Soft Actuator Prototyping:** Built and tested three actuator architectures linking cavity geometry and structural constraint to bending, curling, and blooming motion. ***(2025)***

PUBLICATIONS

Journal Articles

1. **Zhang, B., & You, F. (2020). Research on Vehicle Exterior Interaction Design Based on Automation Acceptance Model. *Journal of Graphics.***
2. Wang, W., **Zhang, B.**, Fu, M., Lin, Z., Pei, J., & Wang, J. (2019). Human-Machine Interface Design of External Displays for Unmanned Logistics Vehicles. *Journal of Graphics.*

Conference Papers

3. Lanzer, M., Babel, F., Yan, F., **Zhang, B.**, You, F., Wang, J., & Baumann, M. (2020). Designing Communication Strategies of Autonomous Vehicles with Pedestrians: An Intercultural Study. *AutomotiveUI 2020.*
4. **Zhang, B., You, F., Yan, F., & Wang, J. (2019). Study on Pedestrian Interaction with Unmanned Logistics Vehicles Based on Cognitive Acceptance. *HHME 2019.***

Books & Chapters

5. **Zhang, B. (Contributing Author).** ***Research on Intelligent Vehicle Interaction Design Methodology.*** **Science Press. ISBN 978-7-03-074373-2.**
6. **Zhang, B. (Chapter Author).** ***Human-Machine Interface for Intelligent Vehicles - Design Methodology and Cognitive Evaluation.*** **Chapter 2.**

WORK EXPERIENCE

SAIC Volkswagen Automotive Co. （FULL TIME ） **Shanghai, China**

HMI Designer in Intelligent Connectivity R&D Department ***Apr. 2023 - Present***

1. Led core interaction-framework design for the next-generation NEX HMI, a China-developed production system designed around localized user needs and driving contexts.
2. Defined the cross-domain HMI framework for the new AUDI intelligent cockpit; the design entered mass production.
3. Led UX/HMI design for the RHP 2.0 automated-parking program, coordinating cross-functional review and contributing to 10 publicly disclosed patents, including five recently granted design patents.

Product Definition Engineer in Autonomous Driving R&D Department *Jul. 2022 - Mar. 2023*

1. Defined end-to-end HMI behavior for the ADAP proof-of-concept project, from interaction framework through simulator validation.
2. **Led interaction design, in-vehicle simulator development, and live-vehicle demonstrations for the RHP 3.0 autonomous-driving concept; the project was selected for the Volkswagen IVET Conference and advanced to production adaptation.**

HMI Designer in Autonomous Driving R&D Department (ERR). *Jul. 2021 - Jun. 2022*

1. **Designed and evaluated HMI concepts for an SVW concept vehicle through prototype evaluation and usability research.**
2. Defined a next-generation ADAS interaction framework and synthesized user research, qualitative analysis, and large-scale test findings.
3. Supported the L4 Autonomous Shuttle X through cross-scenario validation and behavioral analysis of driver-passenger interaction.
4. **Designed interaction strategies for a DMS-based behavioral-alert system, linking driver-state detection to proactive intervention logic.**

BMW GROUP（INTERNSHIP） **Shanghai, China**

Interaction Design Intern, ABK China Skylab *Oct. 2019 – May 2020*

- Contributed to the design of a digital credit system app and the BMW Avatar project; independently led usability studies comparing BMW ID 7 and ID 8 systems.

ADDITIONAL EXPERIENCE & SKILLS

- **Founder & Editor, Beebo HMI Media Platform** (2019 – Present): Founded an independent media brand, publishing 215+ articles and growing a community of ~10,000 followers in the automotive HMI field.
- **Teaching Assistant, Tongji University** (2018 – 2021): Assisted courses in Interface Design, User Research, and Service Design.
- **Research Areas: Human-Robot Interaction | Human-Autonomy Interaction | Embodied Interaction**
- **Methods & Tools: Simulator Studies | Usability Testing | Surveys | Interviews | Eye Tracking | EMG | SPSS | R | Unity | Figma | Arduino | Processing | C/C++**